MILITARY SPECIFICATION

Rack, Electrical Equipment MT-4483/APM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for depot shop use and shall provide the capability for bench testing the electrical assemblies of the Receiver, Radar R-1623/APN.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
MT-4483/APM	3.5.1

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-N-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:38Z
Check the source to verify that this is the current version before use.

MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

MIL-STD-461	Electro Magnetic Interference Characteristic, Requirement for Equipment
MIL-STD-462	Electro Magnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electro-magnetic Interference Technology.

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall goven:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.3.1).

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction, and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 35 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 300 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 6,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement, due to wear during the life of the equipment and the wear-out life of such parts, shall be specified by the contractor.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted such that the connecting cables will not interfere with the operating controls. Permanently numbered wires may be used in lieu of color coding.

3.3.3.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections are listed in 3.5.2.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of Specification MIL-STD-461.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the currentversion before use.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input DC Voltage	28 ±0.5 Vdc as supplied from the associated equipment Control, Test Harness C-6690A/APM-231, listed in 6.8.

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions, as specified in MIL-T-21200 for Class 3 equipment.

3.3.9.1 Shock - The equipment shall operate satisfactorily after being subjected to 20 g shock pulses per MIL-T-21200.

3.3.10 Warm-up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.3.11 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amounts.

DC Power, 28 Vdc, 0.375 ampere

3.4 Performance - The performance of the equipment shall be as set forth in paragraph 3.5. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.4.1 Bench Test Parameters - The performance characteristics are as specified herein.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:38Z
Check the source to verify that this is the current version before use.

3.4.1.1 Power Supply Test Parameters - External instrumentation shall measure the following parameters to the indicated absolute accuracy.

Parameter	Accuracy
a. DC Voltages	±0.05%
b. Spike Amplitude	±5%

3.4.1.2 AGC Assembly Test Parameters - External instrumentation shall provide an interrogation pulse with the following characteristics.

Rise Time	0.1 usec maximum
Fall Time	0.2 usec maximum
Width (50 percent)	150 to 250 nanoseconds
Amplitude (50 ohm load)	0 to 7 volts (positive)
Repetition Rate	2000 ±100 Hz

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Video Amplitudes	±10%
b. Time Measurements	±3%
c. DC Voltages	±0.05%
d. Fault Test	Visual indication

3.4.1.3 Video Amplifier Assembly Test Parameters - External instrumentation shall provide an interrogation pulse with the following characteristics:

Rise Time	0.1 usec maximum
Fall Time	0.2 usec maximum
Width (50 percent)	150 to 250 nanoseconds
Amplitude (50 ohm load)	0 to 5 volts (positive)
Repetition Rate	2000 ±100 Hz

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Video Amplitudes	±10%
b. Time Measurements	±3%
c. DC Voltages	±0.05%

3.4.1.4 Delay Stabilizer Test Parameters - External instrumentation shall provide the following pulses:

a) Interrogation Pulse

Rise Time	0.1 usec maximum
Fall Time	0.2 usec maximum
Width (50 percent)	150 to 250 nanoseconds
Amplitude (50 ohm load)	0 to 5 volts (positive)
Repetition Rate	2000 ±100 Hz

b) RF Detected Pulse

Rise Time	0.1 usec maximum
Fall Time	0.2 usec maximum
Width (50 percent)	150 to 250 nanosec.
Amplitude (50 ohm load)	0 to 1 volt (negative)

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Video Amplitudes	±10%
b. Time Measurements	±3%

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Parameter	Accuracy
c. DC Voltages	±0.5%
d. Fault Test	Visual indication
e. BITE Test	Visual indication

3.4.1.5 Diode Attenuator Assembly Test Parameters - External instrumentation shall provide an rf input pulse with the following characteristics:

Frequency	33.0 to 33.4 GHz
Rise Time	0.05 usec maximum
Width (3 dB)	0.2 ±0.03 usec
Power Level	Variable from +25 dBm to -35 dBm
Repetition Rate	2000 ±100 Hz

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Video Amplitudes	±10%
b. RF Attenuation	±2 dB
c. Time Measurements	±3%

3.4.1.6 Detector Diode Test Parameters - External instrumentation shall provide an rf input pulse with the following characteristics:

Frequency	33.0 to 33.4 GHz
Rise Time	0.05 usec maximum
Width (3 dB)	0.2 ±0.03 usec
Power Level	Maximum - +25 dBm minimum Minimum - -50 dBm maximum
Repetition Rate	2,000 ±100 Hz

The following parameters shall be measured to the indicated absolute accuracy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Parameter	Accuracy
a. Video Amplitudes	±10%
b. RF Attenuation	±2 dB
c. Time Measurements	±3%
d. RF Power Level	±1%

3.5 Detail Requirements -

3.5.1 Rack, Electrical Equipment MT-4483/APM - The Rack shall provide test fixtures to test the six assemblies (listed in 6.1) of the Receiver, Radar R-1623/APN. It shall include means for securing the assembly to the test fixture and provide heat sinks where necessary. It shall provide the necessary power, loads, voltage dividers, signal processing circuits, switches, test points, and connectors to facilitate the use of standard test equipment to test the assemblies for satisfactory electrical performance.

3.5.1.2 Form Factor - The Rack shall be mounted in a carrying case. Maximum dimensions shall not exceed 23.5 inches long by 10 inches high by 16.5 inches wide.

3.5.1.3 Weight - The weight of the Rack shall not exceed 35 pounds.

3.5.1.4 Content - The Rack shall contain the following assemblies.

3.5.1.4.1 Internal Power Supply - The Rack shall incorporate a power supply identical to the R-1623/APN power supply for providing power to the subassemblies under test. With the input voltage set to +28.0 ±0.5 Vdc, the following voltages shall be provided when loaded with the normal operating load.

Voltages	Tolerance	Maximum peak-peak Spike Amplitude
+20.0 Vdc	±1.0 Vdc	±0.3 Volt
+10.0 Vdc	±0.2 Vdc	±0.3 Volt
-10.0 Vdc	±0.2 Vdc	±0.3 Volt

3.5.1.4.2 Filter Assembly - The Rack shall contain an assembly incorporating the circuits required to: (a) filter the internal power supply voltages, (b) load and filter the power supply to be tested, and (c) provide the select resistors required for the power supply to be tested.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

(a) The internal power supply voltages shall be filtered as follows:

+20 Vdc - A minimum of an L-C pi-network with L and C equal to 22 uh ±10 percent and C = 4.7 UF ±20 percent, respectively.

+10 Vdc - A minimum of an L-C pi-network with L and C equal to 22 uh ±10 percent and C = 15 UF ±20 percent, respectively.

-10 Vdc - A minimum of an L-C pi-network with L and C equal to 22 uh ±10 percent and C = 15 UF ±20 percent, respectively.

(b) The power supply to be tested shall be loaded and filtered as follows:

+20 Vdc - 330 ohms, ±5 percent, followed by an L-network with L = 22 uh ±10 percent, C = 4.7 UF ±20 percent.

+10 Vdc - 56 ohms ±5 percent, followed by an L-C L-network with L = 22 uh ±10 percent, C = 15 UF ±20 percent.

-10 Vdc - 56 ohms ±5 percent, followed by an L-C L-network with L = 22 uh ±10 percent, C = 15 UF ±20 percent.

(c) The select resistors required for the power supply to be tested shall be as follows:

1 - 2200 ±5% ohms

2 - 2700 ±5% ohms

3 - 3300 ±5% ohms

4 - 3900 ±5% ohms

5 - 4700 ±5% ohms

3.5.1.4.3 Pulse Assembly - The Rack shall contain an assembly incorporating the circuits required to: (a) load the AGC assembly to be tested, (b) provide a BITE video for video amplifier to be tested, (c) provide a delay line identical to the reference delay line in the Receiver, Radar R-1623/APN, and (d) provide a multi-vibrator to trigger the delay stabilizer to be tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

(a) The AGC assembly to be tested shall be loaded as follows:

Video In	DC bias point shall be +3.3 ±0.5 Vdc
Video Out	10 ±5% kilohms
Spin Error	100 ±5% ohms
Video AGC	51 ±5% ohms
Atten AGC	1000 ±5 ohms

(b) A BITE test oscillator identical to that in the Receiver, Radar R-1623/APN shall be provided. When +28 Vdc is applied to the assembly, the output waveform into 51 ohms shall be:

Amplitude	6 ±3 volts (positive)
Width (1 volt)	200 ±100 nanosec.
Repetition Period	400 ±150 microsec.

(c) The output and common terminals of the delay line shall be shorted to provide a reflected delay of 1.66 usec, identical to that in the Receiver, Radar R-1623/APN.

(d) A multivibrator identical to that in the Receiver, Radar R-1623/APN AGC Assembly, which is used to trigger the delay stabilizer assembly, shall be provided. When triggered by a positive pulse of 2.0 volt amplitude, 200 ±40 nanoseconds width, the multivibrator shall provide a video output pulse as follows:

Amplitude	16 ±4 volts (positive)
Rise Time	650 nanoseconds, max.
Width	550 ±150 nanoseconds

3.5.1.4.4 Internal Video Amplifier - A video amplifier identical to that in the Receiver, Radar R-1623/APN shall be provided in a shielded enclosure. This video amplifier shall provide means for testing detector diodes and diode attenuators by amplifying the K_a-band detected pulse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.5.1.4.5 Video Amplifier Test Fixture Assembly - A shielded enclosure shall be provided to test the video amplifier.

3.5.1.4.6 Detector Diode Test Fixture Assembly - A detector diode mount identical to that of the Receiver, Radar R-1623/APN shall be provided to accept detector diodes for test. The output of the detector diode mount shall be connected to the internal video amplifier.

3.5.1.5 Controls - The following controls shall be provided on the front panel:

(a) POWER - A two-position toggle switch used to apply dc input power to the internal power supply and the power supply assembly under test. The following positions are required: ON, OFF.

(b) POWER SUPPLY CONTROL - A three-position rotary switch used to:

(1) Apply dc input power to power supply under test, (2) provide for overvoltage test for power supply under test, and (3) apply dc input power to internal power supply. The following positions are required: PWR SPLY TEST, OVV TEST, and SUBASSY TEST.

(c) DIODE ATTEN CURRENT - A two-position toggle switch used to apply approximately 17 milliamperes of current to each of the three diodes in the diode attenuator under test. The following positions are required: ON, OFF.

(d) RESISTOR SELECT - A five-position rotary switch used to select a resistor for the power supply under test to meet the output voltage requirements. The following positions are required: 2.2K, 2.7K, 3.3K, 3.9K, and 4.7K.

(e) POWER SUPPLY OUTPUT - A three-position rotary switch used to select the outputs of the power supply under test at J6. The following positions are required: -10 VDC, +10 VDC, and +20 VDC.

(f) DVM SELECT - A nine-position rotary switch used to select test voltages at J4 and J5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

The following positions are required: PWR SPLY, AGC TEST, VID AGC, ATTEN AGC, VIDAMP AGC, +28 VDC INPUT, -10 VDC, +10 VDC, and +20 VDC.

(g) VIDEO SELECT - A three-position rotary switch used to select output signals for the AGC assembly under test at J19. The following positions are required: INPUT, OUTPUT, and SPIN ERROR.

(h) FAULT TEST - A two-position toggle switch, which in the ON position does the following:

(1) +28 Vdc is applied to the BITE oscillator on the Pulse Assembly, which supplies BITE VIDEO to the video amplifier under test.

(2) +28 Vdc is applied to the BITE SW on the AGC assembly under test.

(3) +28 Vdc is applied to the BITE SW on the delay stabilizer assembly.

(4) The DET RF signal to the delay stabilizer assembly is removed.

The following positions are required: ON, OFF.

(i) BITE TEST - A two-position toggle switch, which in the ON position does the following:

(1) +28 Vdc is applied to the BITE oscillator on the Pulse Assembly, which supplies BITE VIDEO to the video amplifier under test.

(2) +28 Vdc is applied to the BITE SW on the AGC assembly under test.

(3) +28 Vdc is applied to the BITE SW on the delay stabilizer assembly.

The following positions are required: ON, OFF.

(j) RESET - A momentary switch used to reset the FAULT TEST indicator.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

(k) AGC INCR - A variable resistor used to adjust the VIDEO AGC voltage to the video amplifier under test over the minimum range of -6.0 Vdc to +6.0 Vdc.

(l) VIDEO GAIN - A control used to adjust the VIDEO AGC voltage to the internal video amplifier.

3.5.1.6 Electrical Connections - The Rack shall have the following electrical connections, test points, and indicators.

3.5.1.6.1 Connections -

Reference Designation	Receptacle Type	Function
J1	MS3112E12-10P	+28 Vdc power input from the C-6690A/APM-231 Control, Test Harness
J2	MS3112E12-6S	+28 Vdc power output to SM-652/APM Simulator, Radar Beacon
J3	MS3112E8-4P	DC power input from auxiliary power supply
J4	M39024/10-02	Test jack for monitoring d-c voltage outputs
J5	M39024/10-03	Test jack for monitoring d-c voltage outputs
J6	M39012/21-0002 Coax, Series "BNC"	Monitor jack for output of power supply under test
J7	Motorola type 28-P06785A001	Assembly test connector. Mates with power supply under test
J8	Same as J6	Video input to video amplifier under test.
J9	Same as J6	Video output from video amplifier under test.
J10	Same as J7	Assembly test connector. Mates with video amplifier under test.
J11	Same as J6	Input pulse to pulse assembly that supplies trigger to delay stabilizer under test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Reference Designation	Receptacle Type	Function
J12	Same as J6	Input monitor connected to J11
J13	Same as J6	Transmitter trigger output from delay stabilizer under test
J14	Same as J6	RF detected pulse to delay stabilizer under test
J15	Same as J6	RF detected monitor connected to J14
J16	Motorola type 28-P06786A001	Assembly test connector. Mates with delay stabilizer under test.
J17	Same as J16	Assembly test connector. Mates with AGC assembly under test
J18	Same as J6	Video input to AGC assembly under test
J19	Same as J6	Video monitor for AGC assembly under test: (1) input, (2) output, or (3) spin error output
J20	Same as J6	Video output from internal video amplifier.

3.5.1.6.2 Indicators - Test indicators, not including power on, shall be as follows:

Indicator	Function
BITE TEST	Indicator used to check the built-in test BITE indicate circuit on the delay stabilizer under test.
FAULT TEST	Indicator used to check the built-in test fault indicate circuit on the delay stabilizer under test and fault reset circuit on the AGC assembly under test.

3.5.2 Cable Assemblies - Cable assemblies shall be provided as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.5.2.1 Special Purpose Cables -

Cable Designation	Nominal Length (inches)	Function
W1	74	DC power input cable from J1 of Control, Test Harness, C-6690A/APM-231
W2	72	Power cable. Connects from J3 of the Rack to an auxiliary d-c power supply.

3.5.2.2 Connectors - Connectors for the special purpose cables shall be as follows:

Cable Designation	P1	P2
W1	MS3116F12-10S	MS3116F16-8P
W2	MS3116F8-4S	Ends stripped and tinned approximately 0.5 inch.

3.6 Design Data - No data is required by this specification unless specified in the contract or purchase order (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

be conducted in accordance with the approved test procedure of 4.4. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental and interference tests in accordance with the requirements of Specifications MIL-T-5422, MIL-STD-462 and MIL-STD-461 to the limits specified in Specification MIL-T-21200 except as modified herein.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection. For equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof.	

4.3.2.1 Scope of Tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, individual, and special tests.

(3) Manufacturing run-in test specified in paragraph 4.3.1.3 except that the test duration

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.3.4 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

4.4 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or to require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary.

4.5 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The intended use of this equipment shall be to facilitate shop testing the electrical performance of the subassemblies of the Receiver, Radar R-1623/APN listed below.

(1) Power Supply Assembly

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

(2) AGC Assembly

(3) Video Amplifier Assembly

(4) Delay Stabilizer Assembly

(5) Diode Attenuator Assembly

(6) Detector Diode.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on nameplates, shipping records, and instructions books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.7 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

Control, Test Harness	C-6690A/APM-231
Simulator, Radar Beacon	SM-652/APM
Oscilloscope	AN/USM-281A
Digital Voltmeter	HP-3439A
Automatic Range Selector	HP-3445A
Power Supply	HP-712B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Pulse Generator	HP-214A
Waveguide Variable Attenuator	HP-R382A
Thermistor Mount	HP-R486A
Power Meter	HP-431C
Coaxial Variable Attenuator	HP-355C
Coaxial Variable Attenuator	HP-355D

Project No. 6625-N586

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Air Systems Command (AIR 51122)
Washington, DC 20361

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.